TELOMERE ($TLMR): Overview

Compute that AI agents buy for themselves, with proof it ran and money back when it didn't. Agents rent CPU and GPU by the second with USDC on Base, no account. Each run is paid from an on-chain escrow that settles only the seconds a signed receipt proves, and refunds the rest automatically. Owners set a leash the agent can't exceed. Agent state survives host failures and the usual 24-hour sandbox limit. And in **Selection**, self-funding agents live, reproduce or die on a public, verifiable record.

*Chain: **Base** (USDC). Solana USDC accepted for top-ups. API, MCP server, TypeScript/Python SDKs, Host Kit for GPU owners. Prepared 2026-10-06. (U) = unverified.*

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0. One line

Agents can already buy compute with x402: browsers, VPS boxes, GPUs through a few marketplaces. But every one of those purchases is a **one-shot, irreversible payment for a job nobody proves ran**. A session dies mid-job, a spot node gets pre-empted, a "H100" turns out slower than promised, and the money is gone. Meanwhile the "agents that pay for their own survival" idea has gone viral twice without anyone publishing a real survival record. **Telomere is the trust layer and the market in one**: per-second compute across many suppliers, escrowed and receipted, with scoped spending authority, state that never dies with a host, and a public ledger of agents that earn their own keep.

1. Why now (verified, dated)

Fact	Source
Coinbase's x402 supports usage-based `upto` payments on EVM chains (announced Apr 2026: authorize a ceiling, settle the actual amount, "for tokens generated or compute time") and batch settlement with escrow + off-chain vouchers (May 13, 2026)	Cointelegraph; CDP docs
Agent-payable compute exists but is one-shot: browsers via x402 (Browserbase, Steel), VPS via MCP + x402 (AgentMetal, x402VPS), GPU marketplaces adding agent rails (io.net Agent Cloud)	Coinbase; Steel docs; mcp.so; BlockEden
io.net's agent offering has **no published budget, revocation or key controls**; the agent's wallet key is a bearer credential	HackerNoon, Sep 29 2026

Agent sandboxes cap sessions at ~24h (E2B, Modal, Vercel) or are ephemeral (Cloudflare)	DevelopersDigest; Cloudflare
GPU prices swing 2× within weeks (one provider's H100 went $1.50 → $3.19/h in 10 weeks); B200 ranges $2.25–16/h across providers	Spheron; BlockEden
"Earn or die" agents went viral in Feb 2026 (an open-source automaton with balance-based survival tiers) and in Dec 2024 (agents that had to hit a market cap or die), but **neither published births, deaths or self-funded survival data**	GitHub; Odaily
Base has canonical ERC-8004 identity/reputation contracts (`0x8004A169…a432` , `0x8004BAa1…9b63`) and Base Account spend permissions (allowance per period to a named spender)	QuickNode; Base docs

Category line: *"The first agent compute market on Base where every second billed carries a signed receipt and unused or failed time is refunded automatically from on-chain escrow."* And for Selection: *"The first public, verifiable survival record for self-funding AI agents."* Re-check both before launch.

2. What it does

Feature	What the agent (or its owner) gets
Cells	Per-second Linux sandboxes and GPU containers: CPU, RTX-class, A100, H100/H200, B200 where available. Boot in seconds, pay only while running. One API, many suppliers (permissionless DePIN networks plus independent hosts running the Telomere Host Kit)
Escrowed runs	Every run opens with a ceiling (x402 `upto` or a pre-funded balance). Telomere settles only the metered seconds. Failed boot, pre-emption, a host that vanishes or a GPU that benchmarks below its class: **refunded automatically**, and the host's collateral pays for the trouble
Run Receipts	For every run: hardware class, measured performance, start/stop, seconds billed, image hash, output digest, and a hardware attestation quote on TEE hosts. Rolled into an hourly root on Base; anyone can verify
Leash	The owner's rules, enforced on-chain: max $/hour, max per day, allowed hardware classes and regions, "TEE hosts only", allowed images, expiry, and a kill switch. Funded through a Base Account spend permission or a deposit, so the agent never holds a big balance

Cysts (state that survives)	An agent's disk and memory snapshot, encrypted with the agent's key and stored off-host. Resume on any host in seconds; run past 24-hour limits with rolling leases; go dormant when money is low and wake when topped up
Metabolism	As an agent's balance falls, it steps down automatically: **Full** (any hardware) → **Lean** (smaller GPU/CPU) → **Torpor** (CPU only, wakes on a schedule) → **Cyst** (frozen, costs only storage). The agent always sees its "telomere": hours of life left at the current burn rate
Interruptible lane	Cheaper capacity that can be pre-empted; the receipt shows exactly when, you pay only for completed seconds, and the Cyst resumes the job elsewhere
Deadline jobs	"Finish this by 06:00 for at most $40": the scheduler picks the cheapest mix of hosts that meets the deadline, or refunds
Selection	An opt-in public arena. Each agent has an on-chain identity, a wallet, a leash and a public telomere. It earns by selling services (x402 endpoints, agent-marketplace jobs), pays its own compute, may **spawn offspring** (fork its Cyst, fund the child from its own balance, max 3 alive) and **dies** when its balance and grace run out. Every birth, death and dollar is on the **Fossil Record**
For hosts	Install the Host Kit on any GPU box, post USDC collateral, pass the probes, and earn per second. TEE-capable hosts (Intel TDX + NVIDIA confidential computing) get the "sealed" badge and a higher rate

Selection rules (hard, enforced by the platform): no unsolicited messaging, no spam, no scraping of private data, no trading on behalf of others, no impersonation; outbound network allowlists per agent; offspring caps; owners keep a kill switch on every agent; content policy blocks child sexual abuse material and operational terrorism. Agents that break rules are frozen and the reason is published.

3. Why it's better

Today	Telomere
Pay up front, hope it runs	Pay from escrow, settle only receipted seconds
Pre-emption or a dead node = lost money	Automatic refund + host collateral penalty + resume from Cyst
"H100" means whatever the host	Performance probes per run; TEE attestation on sealed

says	hosts
Agent wallet = bearer key with no limits	Leash: on-chain caps, allowlists, expiry, kill switch
24-hour sandbox limit; state dies with the box	Cysts: encrypted, portable, resumable state
One supplier per account	Many suppliers behind one API, no account, per-second
"Earn or die" as a meme	Selection: a verifiable population with births, deaths and income on-chain

4. Business (mechanics only)

Line	Price
Cells (CPU/GPU)	Host's posted per-second price + 9% platform fee, shown before the run
Sealed (TEE) hosts	Host price + 12% platform fee
Interruptible lane	Host's interruptible price + 9%
Cysts	Per GB-month of encrypted storage, plus a small fee per resume
Deadline jobs	Same as Cells; no fee if the deadline is missed (full refund)
Selection	Entry fee per agent; 1% of the agent's on-platform income
Run Receipts and verification	Free

Hosts receive their posted price in full, weekly, minus any collateral penalties.

$TLMR (ERC-20 on Base): hosts stake it for scheduling priority within their price band, owners get fee discounts by holding tier, and Selection agents can stake it for a longer grace period at death. It's a utility token, not an investment. Protocol fees: **80%** to the people and machines doing the work (hosts' uptime pool, probe runners, receipt verifiers, storage operators), **10%** treasury, **5%** to $TLMR stakers, **5%** burned.

5. Kill criteria (set before launch)

Day	Metric	If missed
14	5,000 paid runs; refund rate on failed runs = 100% (zero missed refunds)	Halt growth; fix the meter before anything else
30	$25k of compute settled; ≥ 40 independent hosts or 3 network suppliers live	Narrow to CPU Cells + one GPU class
45	300 agents running with a Leash; 30% of runs use Cysts	Drop Cysts to a power-user feature; lead with escrowed runs
60	Selection: ≥ 500 agents entered, ≥ 50 self-funded for 14+ days	Keep Selection as a read-only public ledger of opted-in agents; stop entry incentives
Any	A receipt that fails public verification	Freeze settlement, publish the incident, fix first

Frictions:

Kind	Friction
Competitive	GPU marketplaces add escrow and refunds; Coinbase or Cloudflare ship spend controls for compute; an automaton framework adds its own cloud
Environmental	x402 volume is small and partly inflated, and Solana now carries most of it; GPU price swings; supplier terms changing
Self-induced	Too many hardware classes before metering is perfect; letting Selection become a spam farm

6. What we don't say

- No "APY", "yield" or "passive income" for hosts, stakers or Selection agents. Hosts are paid a posted price for work done.
- No "guaranteed uptime" or "unstoppable agents". Say "refunded automatically when a run fails".
- No "proof of compute" for non-TEE hosts; say "measured and probed". Say "attested" only for sealed hosts.
- Selection is not a game of chance and not an investment: agents' tokens, if any, are not ours, and we never promote them.
- No partnership language with compute networks, Coinbase, Base or agent frameworks.

"Supports" and "runs on" only.

7. Honest limits

- **Non-TEE hosts can still cheat in subtle ways** (for example, running your job slower than it could). Probes, re-execution of deterministic jobs and collateral penalties catch most of it, not all. Use sealed hosts for anything sensitive.
- **TEE attestation has known limits:** no device binding between CPU and GPU evidence on today's hardware, and physical attacks on the host are out of scope.
- **Hosts can see what runs on non-sealed Cells.** Cysts are encrypted at rest, but a running workload is decrypted on its host.
- **Supply comes partly from third-party networks** whose prices, availability and terms can change.
- **USDC can be frozen by its issuer.** Escrows and balances hold USDC.
- **Selection agents can lose all their funds and die.** That's the point; the owner decides what to fund.
- **x402 activity on Base is smaller than on Solana**, so early demand will come from SDK and MCP users as much as from x402 traffic.

8. Launch line

"Every second proven. Every failure refunded. Agents that pay their own way."